

Problem 1c

What I got Wrong: I checked for the wrong assumptions for running Pearson's correlation. Although I correctly did a visual assessment of linearity, I incorrectly ran diagnostics to check the homoscedasticity and normal distribution of the residuals. There was no need for residual analysis. Instead, I needed to check a normal distribution of each variable in the correlation (temperature and elongation rate).

Fixing It: I created a histogram for each variable (temperature and elongation rate) and visually assessed the histogram for normality. The histogram showed that both variables were normally distributed, meeting Pearson's correlation assumptions.

Why This Matters: Checking for residuals was an unnecessary part of the analysis. Furthermore, failing to check for a normal distribution of the variables means I did not know if I could run Pearson's correlation. By checking the normal assumption, I could verify that Pearson's correlation was a valid test to run.

Problem 1e

What I got Wrong: I restated the statistical summary and failed to provide sufficient biological context so that a team studying the health of kelp would understand the study's significance.

Fixing It: I deleted the statistical summary and focused on the biological context, elaborating specifically on how climate change would affect kelp health and the implications that has in kelp conservation.

Why This Matters: Being able to communicate about statistical results in a biological context is important when working with others who are more interested in the implications than the underlying methods. This gives others the test's tangible results and shows an understanding of the biology behind the statistics.

Problem 1f

What I got Wrong: I failed to sufficiently compare Spearman's correlation to Pearson's correlation, including the test results and interpretation.

Fixing It: I explicitly compared the test results for each correlation. I found that the correlations were extremely similar (off by 0.001). From this, the overall interpretation for Spearman's correlation was the same as the interpretation for Pearson's correlation, but I changed the test summary (information in the parenthesis) to reflect the new test.

Why This Matters: Explicitly comparing the tests is important in improving clarity and clearly showing that the two correlation tests, in this scenario, lead to nearly identical results and equivalent interpretations and biological significance.